Chapter 3 Multidimensional Random Variables

Section 3 Independence of Random Variables

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Outline

1. Independence of Random Variables
   - From Conditional Distribution to Independence
   - Definition of Independence
   - Examples
   - Quick Test for Independence
   - General Definition via Events

Motivation: From Conditional to Independence

Previously, we learned about **conditional distributions**. A conditional distribution describes the distribution of one variable **given** the value of another.

Natural Question: Is there a situation where the conditional distribution **does not depend** on the condition at all?

In other words, can knowing the value of Y give us **zero information** about X? If so, this special scenario leads to the concept of **Independence**.

Discrete Case: Deriving Independence

Let (X, Y) be a discrete random vector with joint PMF $p_{ij} = P(X = x_i,\ Y = y_j)$.
The conditional PMF of X given $Y = y_j$ is:

$$P(X = x_i \mid Y = y_j) = \frac{p_{ij}}{p_{\cdot j}}, \quad \text{where } p_{\cdot j} = P(Y = y_j) > 0.$$

If the conditional distribution is independent of the condition $Y = y_j$, we would have:

$$P(X = x_i \mid Y = y_j) = P(X = x_i) \quad \text{for all } i, j.$$

Substituting the conditional formula yields:

$$\frac{p_{ij}}{p_{\cdot j}} = p_{i\cdot} \implies p_{ij} = p_{i\cdot}\, p_{\cdot j} \quad \text{for all } i, j.$$

Conclusion: The joint PMF factors into the product of the marginal PMFs.

Continuous Case: Deriving Independence

Let (X, Y) be continuous with joint density $f(x, y)$ and marginal densities $f_X(x), f_Y(y)$. The conditional density of X given $Y = y$ is:

$$f_{X|Y}(x \mid y) = \frac{f(x, y)}{f_Y(y)}, \quad \text{where } f_Y(y) > 0.$$

If the conditional density does not depend on $Y = y$, we would have:

$$f_{X|Y}(x \mid y) = f_X(x) \quad \text{for all } x, y.$$

Substituting the conditional formula yields:

$$\frac{f(x, y)}{f_Y(y)} = f_X(x) \implies f(x, y) = f_X(x)\, f_Y(y) \quad \text{for all } x, y.$$

Conclusion: The joint density factors into the product of the two marginal densities.

Definition: Independence of Random Variables

Definition: Mutual Independence

Let (X, Y) be a two-dimensional random variable with joint CDF $F(x, y)$, and marginal CDFs $F_X(x)$ and $F_Y(y)$. If for any real numbers x, y, the following holds:

$$F(x, y) = F_X(x)\, F_Y(y),$$

then X and Y are said to be **mutually independent**.

Equivalent conditions for different types:

- **Discrete:** $p_{ij} = p_{i\cdot}\, p_{\cdot j}$ for all i, j.
- **Continuous:** $f(x, y) = f_X(x)\, f_Y(y)$ for almost all x, y.

Example 1: Independent Discrete Variables

Example: Two Independent Shots

A shooter fires twice independently. The probability of hitting the target is $p = \frac{1}{2}$ each time. Let

$$X = \begin{cases} 1, & \text{1st shot hits} \\ 0, & \text{1st shot misses} \end{cases}, \quad Y = \begin{cases} 1, & \text{2nd shot hits} \\ 0, & \text{2nd shot misses} \end{cases}$$

Find the joint PMF and determine if X, Y are independent.

Analysis:
Since the two shots are independent events, $P(X = i, Y = j) = P(X = i)P(Y = j)$. The marginals are $P(X = 1) = P(Y = 1) = \frac{1}{2}$.

Example 1: Solution and Observation

Solution

The joint PMF is:

$$P(X=0,Y=0)=\frac{1}{4},\quad P(X=0,Y=1)=\frac{1}{4},$$

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The table is:

$X\backslash Y$	0	1	$p_{i\cdot}$
0	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{2}$
1	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{2}$
$p_{\cdot j}$	$\frac{1}{2}$	$\frac{1}{2}$	1

Since $p_{ij}=p_{i\cdot}p_{\cdot j}$ holds for all i,j, X and Y are **independent**.

Example 2: Dependent Discrete Variables

Example: Drawing Without Replacement

A bag has 2 red and 3 white balls. Two balls are drawn successively without replacement. Let

$$X = \begin{cases} 1, & \text{1st is red} \\ 0, & \text{1st is white} \end{cases}, \qquad Y = \begin{cases} 1, & \text{2nd is red} \\ 0, & \text{2nd is white} \end{cases}$$

Find the joint PMF and determine if X, Y are independent.

Analysis:

Use the multiplication rule: $P(X = i, Y = j) = P(X = i)P(Y = j \mid X = i)$. For example, $P(X = 1, Y = 1) = \frac{2}{5} \cdot \frac{1}{4} = \frac{1}{10}$.

Example 2: Solution and Observation

Solution

The joint PMF table is:

$X \backslash Y$	0	1	$p_{i\cdot}$
0	$\frac{3}{10}$	$\frac{3}{10}$	$\frac{3}{5}$
1	$\frac{3}{10}$	$\frac{1}{10}$	$\frac{2}{5}$
$p_{\cdot j}$	$\frac{3}{5}$	$\frac{2}{5}$	1

Check $p_{11} = \frac{1}{10}$. However, the product of marginals is:

$$p_{1\cdot}\, p_{\cdot 1} = \frac{2}{5} \cdot \frac{2}{5} = \frac{4}{25}.$$

Since $\frac{1}{10} \neq \frac{4}{25}$, $p_{11} \neq p_{1\cdot} p_{\cdot 1}$. Therefore, X and Y are **not independent**.

Example 3: Independent Continuous Variables

Example: Exponential Density

Let the joint density of (X, Y) be

$$f(x,y) = \begin{cases} e^{-(x+y)}, & x > 0,\ y > 0, \\ 0, & \text{otherwise.} \end{cases}$$

Determine if X and Y are independent.

Example 3: Independent Continuous Variables

Analysis:
First, find the marginal densities by integrating out the other variable.

Solution

For $x > 0$: $f_X(x) = \int_0^{+\infty} e^{-(x+y)} dy = e^{-x} \int_0^{+\infty} e^{-y} dy = e^{-x}$.
For $y > 0$: $f_Y(y) = \int_0^{+\infty} e^{-(x+y)} dx = e^{-y}$.
Thus, $f_X(x) f_Y(y) = e^{-x} e^{-y} = e^{-(x+y)}$ for $x, y > 0$, and 0 otherwise.
Since $f(x, y) = f_X(x) f_Y(y)$ everywhere, X and Y are **independent**.

Example 4: Dependent Continuous Variables (Triangle)

Example: Uniform on a Triangle

Let (X, Y) be uniformly distributed on $D = \{(x, y) : 0 \leq x \leq 1,\ 0 \leq y \leq x\}$, i.e.,

$$f(x, y) = \begin{cases} 2, & 0 \leq x \leq 1,\ 0 \leq y \leq x, \\ 0, & \text{otherwise.} \end{cases}$$

Determine if X and Y are independent.

Example 4: Dependent Continuous Variables (Triangle)

Analysis:
The marginal integrals will have variable bounds because of the triangular support.

Solution

For $0 \leq x \leq 1$: $f_X(x) = \int_0^x 2dy = 2x$.
For $0 \leq y \leq 1$: $f_Y(y) = \int_y^1 2dx = 2(1-y)$.
Then $f_X(x)f_Y(y) = 4x(1-y)$ for $0 \leq x, y \leq 1$.
But $f(x,y) = 2$ on the triangle. They are not equal, so X and Y are **not independent**.

Proposition: Fast Criterion for Independence

Proposition

Let (X, Y) be continuous with joint density $f(x, y)$. If:

1. **Support is a rectangle:** The region where $f(x, y) > 0$ is of the form $a \leq x \leq b,\ c \leq y \leq d$.
2. **Density factors:** On this support, $f(x, y)$ can be written as $g(x)h(y)$.

Then X and Y are independent.

Proof Outline:

For $x \in (a, b)$, $f_X(x) = \int_c^d g(x)h(y)dy = g(x)C_1$.

For $y \in (c, d)$, $f_Y(y) = \int_a^b g(x)h(y)dx = h(y)C_2$.

Since $\iint f(x, y)dxdy = C_1C_2 = 1$, we get $f_X(x)f_Y(y) = g(x)h(y)C_1C_2 = g(x)h(y) = f(x, y)$.

Practical Rules: Look at Support, Then Factor

To check independence for continuous random variables, follow these two steps:

Step 1: Look at the Support Region

Draw the region where $f(x, y) > 0$.

- If the region is a **rectangle** (e.g., $0 < x < 1,\ 0 < y < 2$), proceed to Step 2.
- If the region is **not a rectangle** (e.g., a triangle $0 < y < x < 1$, or bounded by $y = x^2$), **immediately conclude they are NOT independent**.

Step 2: Check Factorization

- If $f(x, y)$ can be factored into $g(x)h(y)$ (e.g., x^2y, $e^{-(x+y)}$), they are **independent**.
- If $f(x, y)$ contains a cross term (e.g., $x + y$, $e^{-(x+y)^2}$), they are **NOT independent**.

Example: Fast Criterion in Action

Example: Determine Independence

Check if X and Y are independent for:

1. $f(x,y) = 6xy^2, \quad 0 < x < 1,\ 0 < y < 1.$
2. $f(x,y) = 2, \quad 0 < x < 1,\ 0 < y < x.$
3. $f(x,y) = e^{-(x+y)}, \quad x > 0,\ y > 0.$
4. $f(x,y) = c(x+y), \quad 0 < x < 1,\ 0 < y < 1.$

Solution using fast rules:

(a) Rectangle support; $6x$ and y^2 factor. $\Rightarrow$ **Independent**.

(b) Triangle support $(0 < y < x < 1)$. $\Rightarrow$ **Not independent**.

(c) Infinite rectangle; $e^{-x}e^{-y}$ factors. $\Rightarrow$ **Independent**.

(d) Rectangle support, but $c(x+y)$ cannot factor into $g(x)h(y)$. $\Rightarrow$ **Not independent**.

General Definition: Using Event Classes

The CDF definition is equivalent to a more fundamental definition based on **event classes**.

Definition: Independence via σ-algebras

Let $\mathcal{F}$ be a probability space. For random variables X and Y, let $\sigma(X)$ be the collection of all events involving only X (its generated σ-algebra), and $\sigma(Y)$ for Y. X and Y are independent if for **any** $A \in \sigma(X)$ and $B \in \sigma(Y)$:

$$P(A \cap B) = P(A)P(B).$$

Connection to CDF Definition

Why is the event-based definition equivalent to $F(x, y) = F_X(x)F_Y(y)$?

Choose special events:

$$A = X^{-1}((-\infty, x]) = \{X \leq x\} \in \sigma(X),$$
$$B = Y^{-1}((-\infty, y]) = \{Y \leq y\} \in \sigma(Y).$$

Independence requires $P(A \cap B) = P(A)P(B)$, which expands to:

$$P(X \leq x,\ Y \leq y) = P(X \leq x)P(Y \leq y) \iff F(x, y) = F_X(x)F_Y(y).$$

Since events of the form $\{X \leq x\}$ generate all of $\sigma(X)$, checking this special case implies independence for all events.

Extension to Multiple Variables

Independence of n Variables

Let $X_1, X_2, \ldots, X_n$ be random variables. They are **mutually independent** if for any $A_i \in \sigma(X_i)$:

$$P(A_1 \cap A_2 \cap \cdots \cap A_n) = P(A_1)P(A_2)\ldots P(A_n).$$

Equivalently, the joint CDF factors:

$$F(x_1, \ldots, x_n) = F_{X_1}(x_1)\ldots F_{X_n}(x_n).$$

Independence of Infinite Families

For a family of random variables $\{X_t : t \in T\}$, we say they are mutually independent if every **finite subset** of them is mutually independent.
This definition bypasses the technical problems of defining infinite intersections of events.

Section Summary

- **Independence Definition:** $F(x, y) = F_X(x)F_Y(y)$.
- **Key Conditions:**
  - Discrete: $p_{ij} = p_{i\cdot}p_{\cdot j}$.
  - Continuous: $f(x, y) = f_X(x)f_Y(y)$.
- **Quick Test:**
  - Support must be a rectangle.
  - Density must factor into $g(x)h(y)$.
- **General Definition:** Independence of events from the σ-algebras: $P(A \cap B) = P(A)P(B)$.